

SIEM Log Analysis Lab: Investigating Failed SSH Authentication Attempts Using Wazuh

Google Cybersecurity Professional Certificate (Coursera)

Scenario

This lab uses a simulated scenario in which I act as a security analyst for the fictional e-commerce company Buttercup Games. The task was to investigate the mail server for failed SSH login attempts where someone tried to authenticate as the root user.

Part A: Getting Familiar with the Dataset

A1. Confirm Ingestion

To confirm the dataset was fully ingested, I set the time range to Absolute (January 1, 2000 to now) and ran a wildcard search (*). The index returned 131,973 hits, well above the 100,000-event benchmark expected for this dataset.



Figure 1: Full wildcard search confirming 131,973 total ingested events.

A2. Identify the Data Sources

The instructions reference a host.keyword field to break down events by source, but this field wasn't present in the Available fields list for this dataset. I used log.file.path instead, since the file path itself contains the source folder name, letting me filter by source.



Figure 2: Isolating the vendor_sales source via log.file.path.

Part B: Investigating the Mail Server

B1. Narrow to the Mail Server

Filtering to log.file.path: *mailsv* returned 10,783 events, all from the mail server's secure.log, which records authentication activity. A sample result already showed a relevant SSH failure: "Failed password for invalid user appserver from 194.8.74.23 port 3351 ssh2."

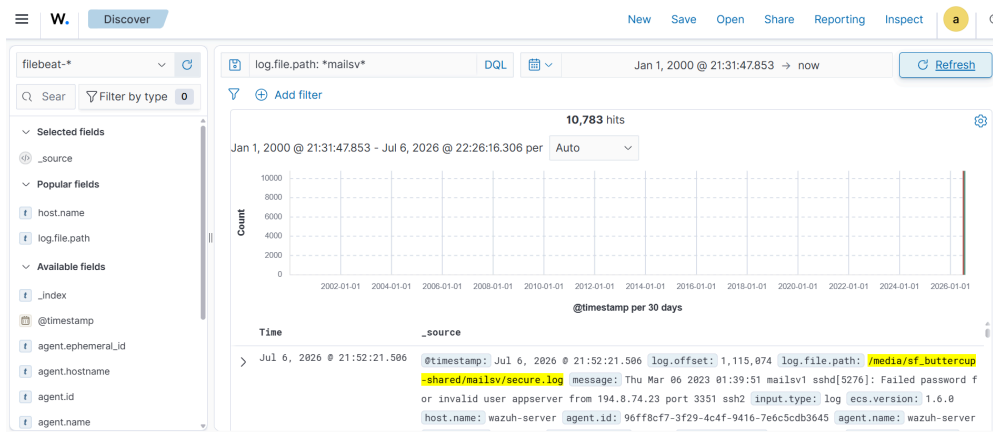


Figure 3: Narrowing to the mail server's secure.log — 10,783 events.

B2. Isolate Failed Root Login Attempts

An early version of this query returned an error due to how the search engine handles wildcards. Rewriting it as an exact phrase search fixed the issue:

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message: "Failed password" AND message: root AND log.file.path: *mailsv*
```

An early attempt at this query returned 722 hits. Re-running the same query a second time, and confirming with the fully-qualified file path as well, consistently returned 384 hits. I used the verified, repeatable number rather than the first result.



Figure 4: Confirmed result — 384 failed root SSH login attempts on the mail server, with search terms highlighted in the matched log line.

Findings

The dataset contained 131,973 total events across five sources.

Narrowing to the mail server's authentication log surfaced 384 failed SSH login attempts where someone attempted to authenticate as root. This volume — hundreds of distinct attempts against the highest-privilege account — is consistent with a sustained brute-force attempt. Possible remediations include checking whether any attempts succeeded, cross-referencing source IPs against threat intelligence, and restricting or disabling SSH root login, with key-based authentication required instead.

Conclusion

In this SIEM investigation lab I verified a dataset, adapted my approach when instructions did not work as expected (missing field), refined a query after hitting a real error (incorrect query syntax), and re-checked a result before reporting it. The 384 failed root login attempts represent an indicator of compromise, suggesting a possible attempt to gain unauthorized access to the mail server and warranting further investigation.